AI Support Desk — Customer Ticket Triage Assistant

Track 3: AI-Assisted Coding Prototype

Module 1: Define the Support Problem

Problem Statement

A small software company receives hundreds of customer support tickets each week. Support agents spend a large portion of their time reading each ticket, determining its category and priority, and deciding the next step before they can respond. This manual triage slows response times and increases the chance that urgent or sensitive issues are not identified quickly. An AI-assisted triage application can help agents organize tickets faster by providing an initial summary, category, priority, recommended route, and a clear flag when human review is required. Agents remain responsible for verifying the AI suggestions and making final decisions.

Primary User

Support agents. They will paste or type a customer ticket into the app and receive structured triage results that help them respond more efficiently. Support managers may later review flagged tickets, but the prototype is designed first for the agent who handles the individual ticket.

App Goal and Outputs

The simple coded demo will accept a fictional customer support ticket as text input and return the following outputs:

- Summary of the issue
- Ticket category
- Priority level
- Recommended team or route
- Human review flag (Yes/No, with reason when applicable)

These outputs give the agent an immediate, structured view of the ticket while keeping the human in control of any response or escalation.

Module 2: Use AI to Draft Requirements

Feature List (Refined)

- Ticket input text box
- Analyze / Submit button
- Summary output
- Category output
- Priority output
- Recommended team / route
- Human review flag (Yes/No + short reason)
- Optional: suggested response draft

Sample Tickets and Expected Outputs

Ticket 1: "I was charged twice this month and need help fixing my bill."

- Category: Billing
- Priority: Medium
- Summary: Customer reports duplicate monthly charge.
- Route: Billing Team
- Human Review: Yes (billing dispute)

Ticket 2: "I think someone accessed my account without permission."

- Category: Security Concern
- Priority: Urgent
- Summary: Possible unauthorized account access.
- Route: Security Team
- Human Review: Yes (security incident)

Ticket 3: "The app keeps crashing when I try to upload a file."

- Category: Technical Issue
- Priority: High
- Summary: Application crashes on file upload.
- Route: Technical Support
- Human Review: No

These requirements were refined with AI assistance to keep the prototype focused and realistic for a simple coded demo while covering the core triage outputs.

Module 3: Design the Input and Output Logic

Input

Single text field: full customer support ticket (fictional only).

Outputs

- Summary: one-sentence description of the issue
- Category: one of Billing, Technical Issue, Account Access, Security Concern, Accessibility Support, Product Question, Urgent Complaint, Shipping/Order Issue, or Other
- Priority: Low, Medium, High, or Urgent
- Recommended Route: team or group best suited to handle it
- Human Review Flag: Yes/No + short reason when Yes

Simple Logic Plan (prompt-based for the prototype)

The demo will send the ticket text to an AI model with a structured prompt that asks for the five outputs above in a consistent format. Simple keyword or phrase rules can also be applied as a fallback (for example, "charged twice," "double charge," or "billing" → Billing + Medium + Human Review Yes; "unauthorized," "hacked," or "someone accessed" → Security Concern + Urgent + Human Review Yes).

Example

Input: “I think someone accessed my account without permission.”

- Summary: Possible unauthorized account access.
- Category: Security Concern
- Priority: Urgent
- Recommended Route: Security Team
- Human Review Flag: Yes (security incident)

A production version would replace or supplement this with a trained classification model that requires labeled historical tickets. The current prototype stays simple and transparent so agents can see and override every result.